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DSA ROADMAP



14 DAY PLAN

DAY 1



Time Complexity and searching

DAY 2



Bubble Sort, Selection and Insertion sort

DAY 3



Recursion

DAY 4



Recursion

DAY 5



Merge Sort Quick Sort

DAY 6



2 Pointers

DAY 7, 8



Sliding Window

DAY 9, 10



Hashmap

DAY 11, 12, 13



Tree

DAY 14



Test



Consistency is the key to mastery. Keep coding, keep improving!



14-Day DSA Roadmap (Placement-Oriented)

Target Companies: Service Based

Class Duration: 3 Hours per Day

Class Structure (Recommended):

- **45 mins:** Theory & Concepts
 - **45 mins:** Dry Runs & Visual Explanation
 - **60 mins:** Coding & Practice
 - **30 mins:** Interview Questions + Homework Discussion
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Day 1 – Time Complexity & Searching

Time Complexity

- What is Time Complexity?
- Why Time Complexity?
- Big O, Big Omega, Big Theta (overview)
- Best, Average & Worst Case
- Rules to Calculate Time Complexity
- Loop Complexity
- Nested Loops
- Logarithmic Complexity
- Space Complexity

Searching

- What is Searching?
- Linear Search
- Binary Search

- Problem Solve
-

Day 2 – Bubble Sort, Selection Sort & Insertion Sort

Bubble Sort

- Working
- Dry Run
- Optimized Bubble Sort
- Time & Space Complexity
- Stability & Adaptiveness

Selection Sort

- Working
- Dry Run
- Finding Minimum Element
- Time & Space Complexity

Insertion Sort

- Working
- Dry Run
- Card Sorting Analogy
- Time & Space Complexity

Comparison

- Bubble vs Selection vs Insertion

Day 3 – Recursion (Basics)

Fundamentals

- What is Recursion?
- Base Case
- Recursive Case
- Function Call Stack
- Recursion Tree
- Stack Overflow

Problems

- Print Numbers (1 to N)
 - Print Numbers (N to 1)
 - Sum of N Numbers
 - Factorial
 - Power Function
 - Fibonacci
 - Reverse String
 - Sum of Digits
-

Day 5 – Merge Sort & Quick Sort

Divide and Conquer

- Concept
- Advantages

Merge Sort

- Merge Function
- Divide Phase
- Merge Phase
- Dry Run
- Complexity
- Stability

Quick Sort

- Pivot Selection
- Partitioning
- Dry Run
- Complexity

- Worst Case
- Random Pivot

Comparison

- Merge Sort vs Quick Sort

Coding Problems

- Merge Two Sorted Arrays
 - Merge Sort
 - Quick Sort
-

Day 6 – Two Pointers

Introduction

- When to Use Two Pointers
- Left & Right Pointer Technique
- Fast & Slow Pointer
- Same Direction vs Opposite Direction

Problems

- Two Sum (Sorted)
- Remove Duplicates
- Move Zeroes
- Reverse Array
- Reverse String
- Valid Palindrome
- Container With Most Water (Concept)

Coding Problems

- Pair Sum
 - Remove Duplicates
 - Move Zeroes
 - Reverse Vowels
 - Merge Sorted Arrays
-

Day 7 – Sliding Window (Fixed Window)

Introduction

- Brute Force vs Sliding Window
- Fixed Size Window
- Window Expansion
- Window Sliding

Problems

- Maximum Sum Subarray of Size K
- Average of Subarray
- First Negative Number
- Maximum Element in Window (Concept)

Coding Problems

- Fixed Window Sum
 - Average
 - Maximum Sum
 - Count Distinct Elements
-

Day 8 – Sliding Window (Variable Window)

Variable Window

- Expand Window
- Shrink Window
- Maintain Conditions

Problems

- Longest Substring Without Repeating Characters
- Maximum Consecutive Ones
- Fruits Into Baskets
- Longest Repeating Character Replacement (Concept)
- Minimum Window Substring (Concept)

Coding Problems

- Longest Unique Substring
 - Longest Ones
 - Smallest Window
 - Character Replacement
-

Day 9 – HashMap (Basics)

Hashing

- What is Hashing?
- Hash Functions
- Collision
- Chaining
- Open Addressing (Overview)

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Problems

- Frequency Count
- Count Duplicates
- Remove Duplicates
- First Unique Character

Coding Problems

- Frequency Map
 - Character Frequency
 - Count Elements
 - Duplicate Detection
-

Problems

- Two Sum
- Subarray Sum Equals K
- Longest Consecutive Sequence
- Majority Element

Day 11 – Trees (Basics)

Introduction

- Tree Terminology
- Root
- Parent
- Child
- Leaf
- Height
- Depth
- Levels

Binary Tree

- Representation
- Traversals
 - Preorder
 - Inorder
 - Postorder

Coding Problems

- Tree Traversals
 - Count Nodes
 - Count Leaves
 - Height of Tree
-

Day 12 – Trees (Intermediate)

Level Order Traversal

- Queue
- BFS

Binary Tree Problems

- Maximum Depth
- Diameter (Concept)
- Balanced Tree
- Left View
- Right View

Coding Problems

- Level Order
 - Height
 - Left View
 - Right View
-

Day 13 – Binary Search Tree (BST)

BST Properties

- Search
- Insert
- Delete (Concept)
- Minimum
- Maximum
- Successor
- Predecessor

Problems

- Validate BST
- Lowest Common Ancestor
- Search in BST

Coding Problems

- Search BST
 - Insert BST
 - Validate BST
 - LCA
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Day 14 – Assessment & Mock Interview

Written Test (60 mins)

MCQs

- Time Complexity
 - Searching
 - Sorting
 - Recursion
 - Two Pointers
 - Sliding Window
 - HashMap
 - Trees
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Coding Test (90 mins)

Choose 4–5 questions such as:

1. Binary Search
2. Bubble/Insertion Sort

3. Recursive Factorial or Fibonacci
 4. Two Sum
 5. Longest Substring Without Repeating Characters
 6. Level Order Traversal
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Interview Round (30 mins)

Viva Questions

- Explain Time Complexity.
 - Why is Binary Search faster than Linear Search?
 - Bubble Sort vs Selection Sort.
 - Why is Insertion Sort good for nearly sorted arrays?
 - Explain recursion with an example.
 - Merge Sort vs Quick Sort.
 - What is the Two Pointer technique?
 - When do we use Sliding Window?
 - Difference between `map` and `unordered_map`.
 - What is a Binary Search Tree?
 - Explain DFS vs BFS.
 - Which DSA pattern would you use for a subarray problem?
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Daily Homework Pattern

Every day, assign:

- **5 MCQs** on the day's concepts.
- **5 Easy coding problems** to reinforce fundamentals.
- **2 Medium coding problems** for interview preparation.
- **1 Revision problem** from the previous day's topic.